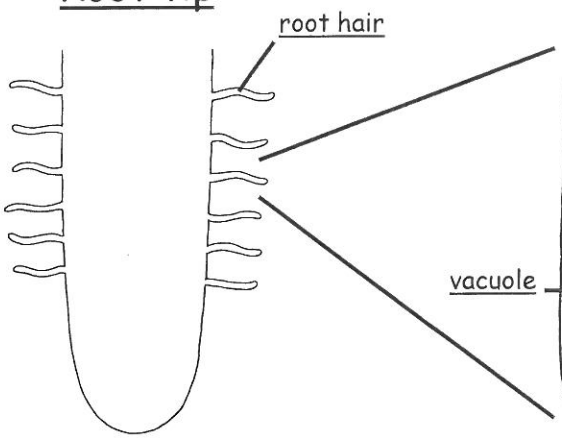
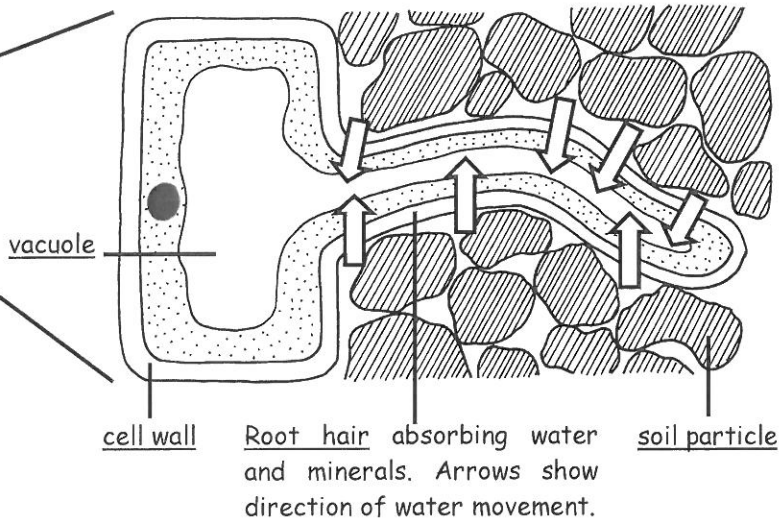
W.S.24. **Plant nutrition.** Name .....

Plants make glucose sugar by the process of photosynthesis. For healthy growth they also need to absorb mineral salts that are dissolved in the soil water. Mineral salts contain elements such as nitrogen, phosphorus and magnesium. Water and mineral salts are absorbed from the soil by the ROOT HAIR CELLS which cover the surface of the root. These cells greatly increase the surface area for absorption.

Root tip




A single root hair cell




| Element | Why it is needed |
|---|---|
| Nitrogen | To make proteins for good growth. |
| Magnesium | To make the green chemical CHLOROPHYLL needed in photosynthesis. |
| Phosphorus | For good root growth. |

Exercise - Complete the missing words in the passage below.

Certain chemical ..... are needed for healthy growth in plants. They are obtained from mineral salts ..... in the soil water. Root ..... cells absorb water and mineral salts which are then carried up the ..... to the leaves. The root hair cells greatly ..... the surface area of the root. If a plant does not have enough ..... it cannot make chlorophyll and its leaves turn yellow. If a plant does not have enough ..... it cannot make proteins and its growth is stunted. Phosphorus is needed for good ..... growth.

**root elements hair magnesium nitrogen dissolved stem increase**

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